Dangling Pointers
Avoid them Strictly!

This article takes off from an earlier one on constant pointers. Here, the focus is on dangling pointers—how they occur and how to prevent them. The article is a great guide for C newbies.

Dangling pointers are those in the C programming language that do not point to valid objects. Pointers are like sharp knives that have tremendous power in C programming. When used properly, they reduce the complexity of the programs to a great extent. But when not used properly, they result in adverse effects and, in the worst case scenario, may crash the system.

In this article, I mainly focus on dangling pointers and what causes them by looking at several different situations in which they occur. I also suggest simple methods to avoid them.

Note: Code snippets provided here are tested with the GCC compiler [gcc version 4.7.3] running under the Linux environment.

Let's now look at three different cases that give rise to dangling pointers.

Case 1: When a function returns the address of the auto variables

Consider the simple code snippet given below (Code 1):

```
#include <stdio.h>

int *fun(void);

int main()
{
        int *int_ptr = fun();
        printf("The value at address %p : %d\n", int_ptr,
   *int_ptr);
```